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# Visual attention and retention effects of a design element in culturally targeted messages: introducing the virtual billboard paradigm

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**ABSTRACT**

Cultural targeting involves designing messages to align with surface or deep-cultural elements with the goal of enhancing message relevance and impact. Methodological challenges remain in experimentally manipulating messages that are congruent with the cultural characteristics of receivers while measuring effects and balancing experimental control and realism; as such, we know very little about mechanisms that explain cultural targeting effects. The current study focuses on the effects of matching messages to self-reported racial identity as a design element of culturally targeted messages. The study used a novel virtual reality-based paradigm in which participants drove along a virtual highway with billboards that portrayed people who either matched or mismatched their racial identity. Using eye-tracking, we observed that participants paid more attention to culturally matched billboards, both in frequency and duration. Results also indicated better recall for matched billboards, suggesting more effective message encoding and storage. This underscores the potential of cultural targeting for health communication and other messaging contexts. More broadly, this study points to new ways for studying communication processes and outcomes with the aid of virtual reality manipulations and biobehavioral measures.

**KEYWORDS**

Billboard paradigm; cultural targeting; eye-tracking; retention; virtual reality

Picture yourself driving down a highway, with billboards stretching along the roadside. One catches your eye – a face that looks familiar and similar to yours. You look closer and may even find yourself thinking about it after your journey ends. Why did this particular billboard grab your attention? What makes some messages resonate more deeply than others? This phenomenon could point to the power of relevance and personal connection in communication. Cultural targeting builds on this principle by aligning persuasive messages with cultural characteristics of their intended audience, thereby presumably enhancing the impact and memorability of messages.

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Nonetheless, questions remain about the outcomes and mechanisms of such messages especially in terms of gains in attention and information retention.

In this paper, we first discuss the role of cultural targeting as a messaging strategy and review the ways it has been used in communication, including providing explanations for its effects. While prior frameworks highlight that cultural targeting can take many forms, we narrow our focus to visual racial matching between message exemplars and recipients as one widely used strategy. Next, we zoom in on the nexus between exposure and reception, or how messages available in peoples' visual communication environments garner attention; an important but underexplored area in the cultural targeting literature. We point out that this exposure-reception nexus, which forms the theoretical foundation of message effects (Slater, 2004), has remained challenging to measure at the individual level, largely due to methodological bottlenecks. We argue that recent advances in virtual reality (VR) technologies with integrated eye-tracking facilities create new opportunities to replicate the effects of billboard advertising with experimental rigor, precision, and ecological validity. Then, the current study is presented in which we display billboard messages presented in a realistic driving environment that either do or do not match participants' self-reported racial identity and examine the effects of this manipulation on attention and subsequent retention. We then discuss the results as well as how our findings advance the theoretical framework underlying the use of cultural targeting. Moreover, we discuss study's novel contributions to studying message reception in realistic, but also highly controlled and experimentally manipulable contexts. This method holds great promise not only for health communication but also political messaging and advertising, as well as mass communication and media effects studies more broadly.

## Background

Messages that align with receiver characteristics tend to resonate more strongly than those that do not (Kreuter et al., 1999; O'Keefe, 2002). Cultural targeting refers to the process of creating messages to address the cultural characteristics of receiver groups. This strategy is widely used in health communication, environmental awareness messaging, advertising, and persuasion at large (Appiah & Liu, 2009; Cho, 2011; Huang & Shen, 2016; Joo & Liu, 2021). The basic ideas behind cultural targeting are connected to the concepts of message targeting and tailoring broadly, which refer to messages designed for groups or individuals, respectively (Noar & Harrington, 2016). Cultural targeting differs from general message targeting in that it focuses on culturally based differences regarding how groups might respond to messages (Lapinski & Oetzel, 2021). Culture, in this context, can be defined as a number of multifaceted characteristics

shared by a group (Kreuter et al., 2003). Cultural targeting is believed to affect how people attend to, receive, and process messages and ultimately their effects.

Cultural targeting can take different forms, and these forms can change the impact of the messages on receivers (Huang & Shen, 2016; Kreuter & McClure, 2004); the forms range from those that target surface-level elements of culture to those that address deep cultural values (Resnicow et al., 1999). Lapinski and Oetzel (2021) expanded the work by Kreuter et al. (2003) to distinguish the core elements of message targeting and tailoring based on the literature on intercultural communication, health communication, and persuasion. One surface-level element is that of *design*; these messages address surface-level elements of culture through the visual, verbal, and non-verbal content and involve "the matching of messages and intervention materials to the observable behavioral and social characteristics of cultural groups" Lapinski and Oetzel (2021), 250; Lapinski et al. (2025). Design elements can include messages which show people who look similar to potential receivers or involve exhibiting symbolic elements of culture (clothing, food, etc.). Researchers can also target more in-depth characteristics such as cultural values or norms (Davis & Resnicow, 2012; Resnicow et al., 1999). Crafting messages that appeal to one's social or cultural identity – including ethnic, racial, gender, or other characteristics (e.g., Joseph et al., 2020) – is noted as one of the ways in which cultural targeting/tailoring can be implemented according to the cultural tailoring content (CTC) framework (Lapinski et al., 2025). In this study, we examine the effects of messages that match participants' self-identified racial identity and therefore use design elements to target the messages.

What can explain or predict the ways in which the elements of the design of messages shape message response? Theory development to explain the possible effects of cultural targeting and tailoring has been sparse, but there are several possible explanations for how and why design elements shape attention to messages and retention of the information in those messages. First, design elements of messages that show, for example, people who look like audience members may function to attract our attention based on principles of similarity such as mere exposure (Zajonc, 1968), liking or attraction (Simons et al., 1970), or the potential biological connections to people who look like us (Bailenson et al., 2008). That is, we tend to positively evaluate things and people that are familiar, like people who are more similar to us, and find them more socially attractive. We may even be biologically predisposed to think positively of people who are similar to us because it is evolutionarily functional. In each of these cases, visual attention should more likely to be focused toward objects and people that are more familiar or similar to ourselves relative to those that are less similar to us; when messages contain images of culturally similar others, we will be visually attracted to such images, resulting in more fixations or longer gaze times.

Design elements that depict members of an ingroup with which one identifies might also operate at a deeper level. For example, messages that show people who look like the receivers might function to bring one's social identity into focus. Social identity theory (SIT) explains how intergroup relations are perceived and unfold within interpersonal processes (Hornsey, 2008; Tajfel, 1974). SIT argues that people are inclined to establish social identity by making distinctions between their ingroup and out-group, and by reinforcing them by attaching a positive image to their ingroup (Tajfel, 1978; Tajfel & Turner, 1986). Self-categorization theory (SCT) is related to SIT and postulates that individuals enhance their identity as a group when they are exposed to situations where their group membership becomes salient, whether it happens transiently or through a prolonged period of time and use it to inform their decision-making (Turner et al., 1987). Aligned with these theoretical assumptions, ingroup identification has been noted as a significant factor that affects message attention, recall, and subsequent persuasion effects, as exemplars of ingroup presented in messages are likely to attract more attention compared to outgroup members (Hale, 2022; Hogg & Reid, 2006; Hogg & Smith, 2007; Wyer, 2010).

In short, we may infer that design features of messages can harness basic psycho-social processes and as such can garner attention and improve retention. Empirical data suggest that messages that match a recipient's cultural features, such as race, ethnicity, or values, are processed preferentially and have a larger effect on behavioral intent and behaviors (Huang & Shen, 2016) than untargeted messages, but questions remain regarding the mechanisms by which this advantage is achieved (Hawkins et al., 2008; Lapinski & Oetzel, 2021).

### *Examining effects of cultural targeting: old challenges and new opportunities*

Although targeting, including cultural targeting, is one of the message strategies used in both communication research and practice, many questions remain regarding its underlying mechanisms in part because research in public health, where cultural targeting is most often studied, typically target along multiple dimensions (Torres-Ruiz et al., 2018). Studying the effects of culturally targeted messages is complicated as it requires experimental designs in which messages are adapted to match specific individual or group characteristics, as well as measuring the minutiae of message reception mechanisms. Moreover, as with all communication and message effects research, it is a challenge to balance experimental control and ecological validity (e.g., Levine, 2018; Miller et al., 2019).

Here, we present a new solution to these challenges that leverages a recently described VR paradigm (Schmälzle et al., 2023). This paradigm uses VR to create a message reception context, i.e., an artificial, but realistic and naturally

navigable environment. In this case, participants experience a virtual drive along a realistic highway with billboards. Specifically, they enter a 3D-photogrammetry version of a real-world highway (Highway 50 near Cold Springs, Nevada, USA) that was digitized by the Nevada Department of Transportation. They can drive through this environment much like in a driving simulator or an immersive computer game, but with the difference that they are actually fully immersed in the spatial surroundings.

One of the main benefits of using VR technology in message effects research is its realism and the potential to manipulate theoretical factors in situ (Ahn, 2015; Martingano & Persky, 2021). Specifically, VR-immersed participants can drive along the highway at their own pace, look around, and explore their surroundings, which include various billboards that are typical of the US highway system. Traditionally, billboards as message channels offer high visibility and media efficiency by delivering exposures at relatively low cost per impressions, particularly in high-traffic areas (e.g., Taylor, 2010; Wilson, 2023). This line of inquiry is also timely as the Out-of-Home (OOH) advertising market is consistently growing fueled by rapidly evolving technologies such as digital billboards and displays (Out of Home Advertising Association of America [OAAA], 2024; Statista, 2023). Given this trend toward more dynamic and technologically advanced message delivery in public spaces, understanding the cognitive and affective factors that drive attention and retention is increasingly critical.

Unbeknownst to participants in the current study, who encounter these billboards as if they were just there, the messages that are placed on the billboard stands can be experimentally manipulated – a little bit like in the famous movies such as Niccol and Weir (1998), Spielberg (2002), or Wachowski et al. (1999). This allows, for instance, the display of messages that either match or don't match participants' racial identity (or any other message characteristic that is of theoretical interest to experimenters). In addition to offering a way to create artificial but realistic environments that can be experimentally manipulated, another benefit of VR is the enormous measurement potential it affords. Specifically, via VR-integrated eye trackers, it becomes possible to keep track of where in their visual field participants are looking and relate this information about their overt visual attention to the information content in the environment, such as the specific billboard and whether it is targeted/matched or non-targeted /unmatched. Accurately measuring message exposure is crucial to persuasive message studies across diverse contexts (Morris et al., 2009; Slater, 2004). VR-integrated eye tracking specifically allows to study how participants incidentally attend to (i.e., look at) messages embedded in complex and dynamic environments that are actively navigated by the user, as opposed to screen-based eye-tracking paradigms in which message exposure is typically forced. It is important to note that the advantage of VR-

integrated eye-tracking in terms of ecological validity is particularly pronounced in contexts such as billboards or outdoor health messaging, while the use of screen-based eye-tracking might be well suitable for study of health warning messages in magazines and online media (see for instance Liu et al., 2021). As such, VR is a methodological innovation for message effects research that promotes method-theory synergy because it allows researchers to manipulate features of messages as well as the environments in which they are embedded, and to rigorously measure relevant process variables (as opposed to retrospective self-report or aggregate metrics of exposure) with high levels of ecological validity.

### *The current study and hypotheses*

The VR-based approach described above makes it possible to focus on the nexus between incidental exposure to messages (by letting participants drive and look freely), message reception (by recording the number of fixations and total gaze duration toward the billboards), and subsequent effects (by measuring which of the billboards they remember). As participants drive down the highway, they pass by culturally targeted billboards, including those matched or not matched to participants' self-identified race. This precise manipulation allowed us to systematically investigate the impact of culturally targeted billboard advertising on potential consumer attention and memory using eye-tracking technology within a VR environment. The billboards feature a variety of health-related topics that are typical of highway billboards, such as buckling-up, distracted driving, drunk driving, as well as smoking, alcohol, and other health, risk, and safety-related topics. Using VR, we can measure whether, for how long, and how often they look at billboards they pass, which quantifies their overt visual attention toward each message. Unbeknownst to participants, the VR environment is pre-programmed to adapt to the participants' cultural identity: the specific billboards that are displayed are manipulated to match/mismatch with the participant/group characteristic, which is the key rationale behind cultural targeting. Once participants arrive at the end of the highway, we test their incidental memory for the messages – first via an unannounced free recall test and then via a recognition memory test.

Drawing upon the theoretical framework outlined previously, we propose that the mechanisms underlying the effects of culturally targeted exemplars in this study can be explained from aforementioned research on social identification and source similarity (Hayashi et al., 2018; McQueen et al., 2011; Tharp-Taylor et al., 2012; Turner & Reynolds, 2012).

Previous literature indicates that culturally targeted messages, by resonating more deeply with the viewer's identity, enhance attention, processing depth, and recall (Appiah & Liu, 2009; Torres & Briggs, 2007). Through exposure to messages that highlight one's racial identity (i.e.,

presenting those that look similar), it is more likely that culturally targeted messages will activate one's ingroup heuristics, which will, as a result, attract greater attention. This, in turn, would lead to greater retention abilities such as recall and recognition. Thus, based on the theory and rationale laid out above, we formulate the following hypotheses and research questions:

**H1.** *Culturally targeted messages will command greater visual attention – a) longer gaze duration and b) more fixations – compared to non-targeted messages (i.e., mismatching messages and neutral messages).*

**H2.** *Culturally targeted messages will result in better retention – a) free recall and b) recognition – compared to non-targeted messages (i.e., mismatching messages and neutral messages).*

**H3.** *Overt visual attention – a) fixations and b) gaze durations – will generally impact retention (i.e., a) free recall and b) recognition).*

**RQ1.** *How do the individual billboard messages or specific topics impact viewers' attention and retention, and how do personal characteristics influence these outcomes?*

## Method

### *Participants*

Participants were recruited from a university participant pool and via word of mouth. The eligibility criteria included 1) those over age 18 and 2) those who had a normal or adjusted vision. The sample size was determined *a priori* and set to 40. This was done following prior research, which demonstrated that this sample size is sufficient to detect effects of fixations on memory outcomes and differences between directed attention and free viewing (Schmälzle et al., 2023). Moreover, power analysis suggests that for a within-subject comparison, samples of around 30 are adequately powered to detect medium and larger effects.[1]The study was approved by the university institutional review board, and all participants provided written informed consent and were compensated for their participation. We collected data from 40 participants and immediately replaced participants who reported blurry vision or experienced other issues ($n = 2$). We excluded one participant who identified as neither White/Caucasian nor Black/African American. We also excluded one outlier after an exploratory analysis of the eye-tracking metrics whose data was abnormal from the rest of the data. Therefore, the final sample size was 38[2]($M_{Age} = 22$, $SD_{Age} = 6.2$; 21 females).

## Materials and equipment

*Equipment and VR-based Eye-Tracking*. The experiment was run in a VR research lab on a GPU-powered gaming PC. We used the HP Reverb G2 Omnicept VR headset with eye-tracking capabilities (Tobii AB, 120 Hz sampling rate). A Python-based VR research platform (Vizard, Worldvision Inc.) was used to program the experiment, using the Sightlab VR package to capture eye-tracking in the VR environment. The environment used for driving was a virtual 3D-photogrammetric model of a highway in Nevada that has also been used in prior research (Schmälzle et al., 2023), in which VR-immersed participants could drive down while freely looking at billboard messages that were placed along the road (see Figure 1).

*Billboard Messages*. Participants were exposed to a total of 20 billboard messages in their VR driving experience including 15 health messages and 5 commercial advertisements as fillers. 15 health messages covered a diverse range of topics, including binge drinking, underage drinking, having healthy eating habits, smoking, cannabis use, avoiding texting while driving, etc. Among the 15 health-related billboards, five billboards presented exemplars that matched their self-reported racial identity (i.e., culturally targeted), five

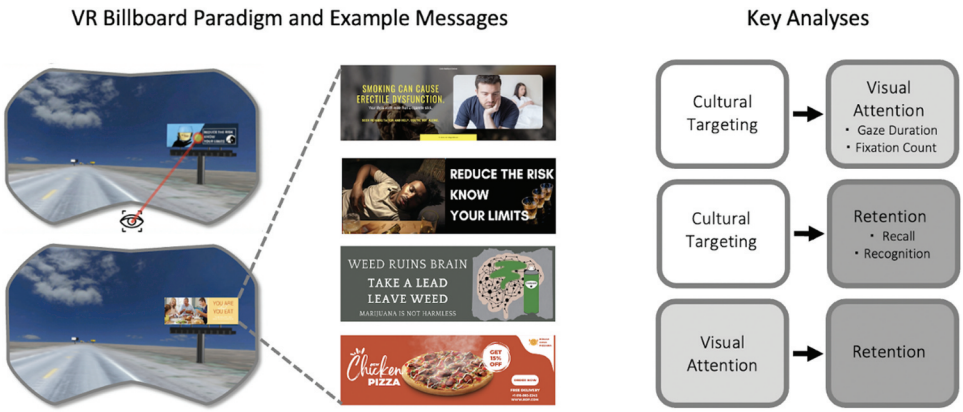



**Figure 1.** Illustration of the VR billboard paradigm and its adaptation in the current study to examine the effects of cultural targeting. A realistic photogrammetry-based model of a highway (highway US50 in Nevada) serves as the virtual environment. Via a VR-integrated eye tracker, we can objectively track whether, for how long, and how often participants look at billboards they pass during a simulated drive. Unbeknownst to participants, the billboards are manipulated to either match or mismatch participants' self-reported racial identity. For instance, in the depicted exemplar, there are two versions depicting a family dinner among either a family of white/Caucasian Americans or black/African Americans. In essence, each billboard exists in multiple versions (e.g., for buckling up, smoking prevention, etc.) that attempt to maximize the overlap between participants' cultural background and the way the message depicts models with a matching background. Our experimental hypotheses then focus on whether the matched/unmatched billboards impact attention (gaze behavior) and whether matching impacts retention (memory measures). We also test the general notion that attention should predict memory, i.e. that unattended messages are less remembered, and that more attention predicts better memory.

presented exemplars that did not match their racial identity (i.e., culturally non-targeted), and five additional health-related billboards did not contain people. For the 10 messages that were assigned to match vs. non-match the participant's self-reported race, we created 2 versions of each – one with a White character and another one with a Black character. The placement, colors, and font of the texts and images were held constant throughout to keep them highly comparable in appearance, with the main variation being the portrayed race of the exemplars. The assignment of the matching versions of billboards was completely balanced between self-reported Black and White participants in terms of which of the two versions was assigned and the specific order in which they appeared. The images of the billboards were adapted from image search using Google and only used when they were free of use in terms of copyrights. Other images were generated using the Midjourney.com generative AI service.

We used a custom-made Python program to create a pseudorandomized assignment of billboards to the matching and non-matching conditions so that across the 40 participants, all billboard versions were used equally often and balanced across participant groups as well as roadside positions.

### *Experimental procedure and conditions*

The procedure for this study comprised an arrival phase, a VR setup and a short demo phase, the main experimental highway drive, and a post-drive interview that included an unannounced memory test and survey questions. In brief, the participants came into the lab in person, signed the consent form, and completed a brief vision test before putting on the VR headset.

Once they put on the VR headset, the researchers executed the VR eye-tracking calibration routine and then allowed the participants to practice navigating the demo virtual highway environment without the billboards. Next, the participants drove down the full virtual highway with the billboards. Because the highway was perfectly straight, no steering was required and participants drove at a constant speed past all billboards. While prior driving experience was not required to participate in the study, all participants mentioned they had been driving before. VR driving itself required minimal effort – all participants needed to do is press the trigger button on the controller to accelerate the vehicle. The full drive took about 10 minutes.

During the post-experiment phase, the participants were given a sudoku puzzle for 2 minutes to clear out their immediate short-term memory contents and prevent recency effects. Then, the researcher(s) conducted a brief structured interview, during which they asked the participants about the length of the drive, their general experience in VR, and every billboard-related factor they recalled (e.g., "Now, please tell us everything you remember about the billboards you passed."). Next, they partook in the recognition test, marking

the billboards that they recognized (shown on a paper survey along with multiple distractors). Finally, the participants completed an online survey via Qualtrics, which included questions about demographics and VR experience.

### Measures

The main measured variables in this study comprise eye-tracking data (whether/how long each billboard was looked at) and memory outcome measures (whether a billboard was later recalled or recognized; Kim & Southwell, 2017). The entire billboard was set as a single area-of-interest (AOI) for the eye-tracking fixation data. Specifically, using the VR-integrated eye-tracker, we determined the total number of fixations and the overall duration of gaze toward the messages. As stated above, eye gaze was tracked unobtrusively while participants underwent the VR driving experience. The Sightlab VR experiment software sampled gaze locations at a rate of 120 Hz and used the ID-T (Salvucci & Goldberg, 2000) algorithm for fixation identification. The fixation threshold was set to 250 ms, in accordance with research on natural scene perception and oculomotor control (e.g., Henderson & Hollingworth, 1999).

To assess memory, we used a free recall task administered at the end of the highway drive right after participants removed the VR headset and completed a short interview. Recognition memory was then tested via a pen and paper survey using a digital tablet that included all experimental billboard images along with four distractor images to gauge potential guessing strategies. Every participant received a recognition datasheet with 24 billboard messages on it (randomly ordered) and, participants had to indicate whether or not they had seen the respective billboard message. The same post-experimental survey also included measures of participants' racial/cultural identity, demographic variables, as well as potential VR-related symptoms (Figure 2), and perceptions of spatial presence (adapted from Kim, 2023; Schmälzle et al., 2023; Souza et al., 2021).

### Data analysis

The main variables of our experiment were i) participants' self-reported racial identity (person variable), ii) the content of the billboard (matching or non-matching or not relevant to racial identity), iii) how often participants looked at (fixated) on each billboard message and for how long they looked at it in total (self-defined behavioral variables), and iv) whether participants recalled/remembered the billboard (also self-defined). Importantly, while participant identity and billboard content were manipulated experimental variables, the viewing behavior and the message memory are self-defined behavioral variables (i.e., different participants decided to look at specific messages and not others, and different participants remembered different billboards).

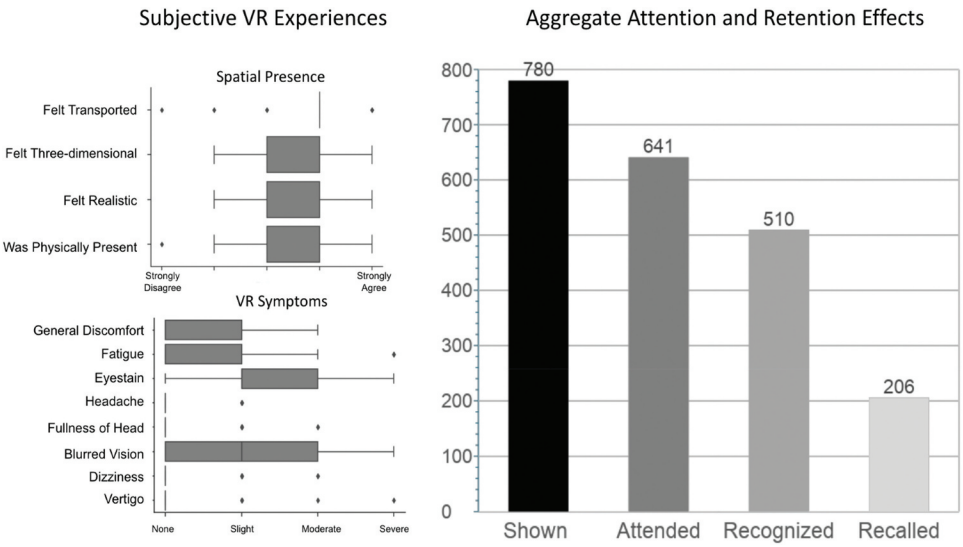



**Figure 2.** Left: subjective reports about their experiences during the VR-mediated highway drive confirm that participants felt spatially present in the highway environment and experienced little to no symptoms. Right: aggregate effects of exposure to messages on attention and retention. Out of the 780 hypothetical exposures (38 participants all passing 20 billboards: $38 \times 20 = 760$), 641 billboards were explicitly looked at (fixated at least 1*), and 510 were recognized and 206 recalled in the post-drive interview. These data confirm the classical arguments of McGuire and advertising researchers more broadly who argue for a multiplicative decay-factor of ca. .7 along the exposure-to-attention-to-retention pathway.

Eye movement behavior and memory data were analyzed using Python code. We document the analysis and provide code in the study's online repository at https://github.com/nomcomm/vr_billboard_t.git. Specifically, we read in the output file from the VR-eye-tracking, which contained information about which billboards a participant had fixated, with the information from the post-study interview and online survey, where memory was assessed for all billboards. Thus, the analytical dataset comprises a table for each participant, containing information about which billboard and which version (e.g., a "don't text and drive" billboard featuring African American models) was displayed, whether it was looked at, and whether it was subsequently remembered. Based on this dataset, we then collapsed data based on whether or not the message matched the participant's racial identity and computed dependent variables in the resulting cells (e.g., share of fixated billboards in the matched vs. unmatched, share of recalled and remembered billboards in these conditions).

In the analysis, we first compared the effect of message-matching on attention (fixation count and gaze length) and retention (recall and recognition rate) separately and then combined both eye-tracking and memory outcomes in a joint analysis.

## Results

### Participants' experiences and behavior during the VR highway drive

First, to examine participants' experiences of spatial presence and the effects of VR during the drive, we analyzed their open-ended responses from the post-drive interview and survey answers. Overall, participants expressed that they felt immersed in the environment as if they were actually driving on the highway, and several participants described their driving experience as calm, and relaxing. Confirming these impressions, the participants reported perceptions of being spatially present in the environment slightly above the scale midpoint ($M = 3.62$, $SD = 0.65$; range 1–5) and relatively low levels of dizziness or discomfort ($M = 1.59$, $SD = 0.62$; range 1–4) (Figure 2).

Next, we turn to participants' behavior during the drive, examining whether they looked at the billboards. We found that many participants looked at the majority of the billboards they passed ($M = 16.4$, $SD = 4$; out of 20 possible), with only 10 participants looking at less than 15 of the 20 they passed, which could be due to participant inattention or measurement issues. In any case, these results confirm that participants scanned the dynamic visual environment they found themselves in and paid at least basic attention to their surroundings. Across the 38 participants, each of whom passed 20 billboards (i.e., $38 \times 20 = 760$ "assumed" exposures), we obtained fixations on 641 billboards, or about 84% ("attended" exposure, cf. Southwell et al., 2002). Overall, participants recognized 510 billboards (67% of all possibilities and 80% of the attended ones), and they freely recalled 206, which amounts to about 27% (of the 760 presented) or 30% (of the 641 attended) billboards. Broadly, this funnel-shaped reduction from exposure to a minimal message effect – defined here as being able to actively retrieve the message from memory – is compatible with McGuire's message effects framework (McGuire, 2001) and results from the advertising literature more broadly (Rossiter & Bellman, 2005). Moreover, the survey results demonstrate that the participants immersed in the driving experience, were not adversely affected by the VR technology, and reported a general sense of satisfaction and enjoyment (see Figure 2).

### Effect of cultural targeting on overt visual attention

For hypotheses testing, we first investigated the effect of culturally targeted/non-targeted messages on visual attention, which encompasses a) gaze duration and b) fixation count (H1) (see Table 1 for descriptives).

#### Gaze duration

We first examined whether matching affected the gaze duration of the billboards. Thus, for every participant, we computed one gaze duration measure for the matched billboards and a corresponding measure for the non-matched

**Table 1.** Means and standard deviations for attention and retention measures across message conditions.

| | | *Conditions* | | | | | |
|---|---|---|---|---|---|---|---|
| | | Targeted | | Non-targeted | | Neutral | |
| | *Measures* | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| Visual Overt Attention | Gaze Duration | 3.074 | 1.493 | 2.851 | 1.422 | 3.135 | 1.246 |
| | Fixation Count | 2.457 | 1.043 | 2.365 | 0.959 | 2.542 | 0.933 |
| Retention | Recognition | 0.788 | 0.262 | 0.820 | 0.259 | 0.710 | 0.267 |
| | Recall | 0.356 | 0.249 | 0.274 | 0.214 | 0.297 | 0.171 |

*Note*. Gaze duration is calculated in seconds, fixation count denotes the number of times a billboard was viewed at with at least one fixation. Retention scores are reported as rate indices.

ones, and we compared these measures at the group level via a paired samples *t*-test. The results are illustrated in Figure 3 (top row). On average, participants looked at the messages that matched their cultural identity for about 3.074 seconds ($mean_{GazeDuration:\ matched} = 3.074$, $SD = 1.493$) while they gazed at the unmatched messages for only about 2.851 seconds ($mean_{GazeDuration:\ unmatched} = 2.851$, $SD = 1.422$). Although this suggests a nominally longer gaze duration toward matched messages, this difference was not statistically significant ($t(37) = 1.213$, $p = .116$).

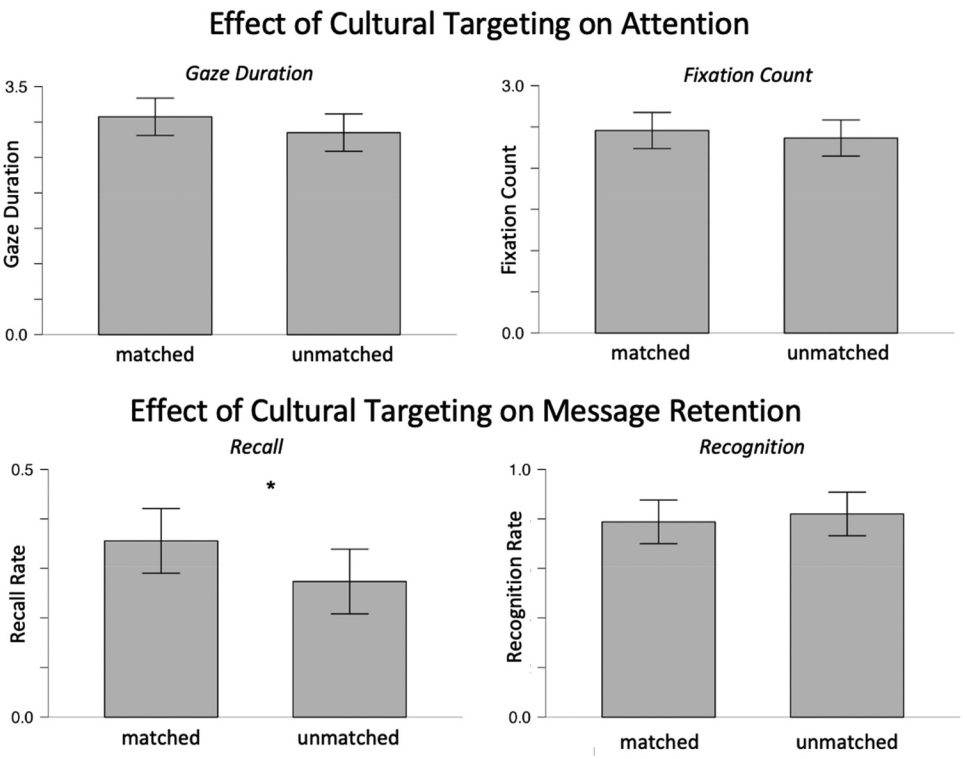



**Figure 3.** Top row: effect of matching on attention: fixation count and gaze duration. Bottom row: effect of matching on retention: recall and recognition.

Furthermore, we assessed gaze duration toward neutral health messages (i.e., messages without human exemplars), which was, on average, about 3 seconds ($mean_{GazeDuration:\ neutral\ health\ messages} = 3.135$, $SD = 1.246$).

### Fixation count

The gaze duration measure above represents the total time spent inspecting a billboard. An alternative metric of visual attention to the billboards is the number of fixations toward a given billboard, meaning that a person might fixate on a billboard, then look somewhere else, but then revisit it for deeper re-inspection. As for the gaze duration measure, we found a slightly higher number of fixations toward matched compared to unmatched messages ($mean_{FixationCount:\ matched\ messages} = 2.457$, $SD = 1.043$; $mean_{FixationCount:\ unmatched\ messages} = 2.365$, $SD = 0.959$), but the statistical comparison was also not significant; $t(37) = 0.601$, $p = .276$).

Again, we also assessed the number of fixations toward non-exemplar messages (i.e., neutral billboards about health-related topics without characters), finding them to lie at 2.542 fixations on average ($mean_{FixationCount:\ neutral\ health\ messages} = 2.542$, $SD = 0.933$).

Overall, although the pattern of results aligns with our prediction of more attention toward matched messages, the data are not consistent with H1.

## Effect of cultural targeting on retention

For H2, we anticipated that culturally targeted messages would garner better retention, which includes a) recognition and b) recall, compared to non-targeted messages (Table 1).

### Recall

As expected, results showed that culturally targeted messages had a greater recall rate ($mean_{Recall\ Rate:\ matched\ messages} = 0.356$, $SD = 0.249$), compared to exposure to non-targeted messages ($mean_{Recall\ Rate:\ unmatched\ messages} = 0.274$, $SD = 0.214$). This difference was statistically significant ($t(37) = 1.8$, $p = .040$). For comparison, the recall rate for neutral health messages was again approximately similar to the unmatched messages, with a $mean_{Recall\ Rate:\ neutral\ health\ messages} = 0.297$ ($SD = 0.171$). These results support H2a, which predicted better recall for targeted messages relative to untargeted messages.

### Recognition

Running the same set of analyses for the data from the recognition task revealed the following results: As expected, the recognition rates are considerably higher than recall, with values of about 80% for recognition compared to 30% for free recall. Levels of recognition memory did not significantly differ between targeted and non-targeted messages

($mean_{Recognition\ Rate:\ matched\ messages} = 0.788$, $SD = 0.262$; $mean_{Recognition\ Rate:\ unmatched\ messages} = 0.820$, $SD = 0.259$; $t(37) = -0.52$, $p = .697$). As a comparison, the neutral health messages were recognized about 70% of the time ($mean_{Recognition\ Rate:\ neutral\ health\ messages} = 0.71$, $SD = 0.267$).

### Impact of overt attention on retention

Furthermore, we also examine the impact of attention (looking at a billboard) on retention (i.e., recognition and recall) (H3). It should be noted, however, a methodological complication arose from our free-viewing instruction, which had the effect that a majority of participants fixated on all or most of the billboards (e.g., 11 participants had looked at all 20 billboards) – and those subjects would then logically not be able to contribute any retention measures for non-attended billboards. Furthermore, participants who looked at most of the billboards (e.g., 19, 18, 17 out of 20 – ignoring 1,2, or 3 billboards, respectively) will have a strong imbalance in the numbers of items on which memory estimates are based (and thus inflated variability). To overcome this, we used generalized linear mixed models and also ran a series of confirmatory analyses using conventional statistics, using only participants who had not looked at all billboards. Regardless of these complexities, the data supports the notion that billboards that were more attended were more likely to be retained.

#### Recall

For the effect of the number of fixations on recall, we find a significant effect ($p < .05$, $\chi^2 = 4.678$), suggesting that more fixations on billboards lead to better recall. The same was the case using gaze duration, with billboards that were looked at longer having a much higher likelihood of recall ($p < .01$, $\chi^2 = 7.615$).

#### Recognition

Parallel results were obtained for recognition. Billboards that were more often fixated were more likely to be recognized ($p < .001$, $\chi^2 = 21.895$), and the same was the case for the gaze duration ($p < .001$, $\chi^2 = 40.902$).

### Exploratory and supplementary analyses

In addition to testing our main *a-priori* hypotheses, we conducted exploratory analyses to elucidate the contribution of message-level characteristics (i.e., the content and topic of individual billboards) and participant-level attributes to viewer attention and recall (RQ1). First, given the longstanding debates around single- vs. multiple-message designs (e.g., Clark, 1973; Jackson & Jacobs, 1983; O'Keefe, 2024), we zoomed in on gaze and memory results for individual messages. Second, we examined each racial subgroup of participants (i.e., Black/African American and White/Caucasian participants).

### *By-message (by-billboard) analyses and supplementary statistics*

The main results reported above were based on conventional *t*-tests or ANOVA-style analyses in which data (e.g., gaze duration, fixation count, recall rate, recognition rate) were collapsed across subjects and stimuli. While these analyses are fairly straightforward and easy to interpret, there is a recent trend toward models that account for the nested structure of items/conditions/participants and the associated variances. To also examine the data this way, we ran mixed effects models (also see Supplementary Data for plots). These models confirmed the conventional results obtained for H1 (effect of matching on gaze duration: $F_{matching\ status}$ (1, 262.91) = 1.464, $p = .227$; effect of matching on fixation count: $F_{maching\ status}$ (1, 279.48) = 0.381, $p = .538$). For H2, the nonsignificant effect of matching on recognition was also seen ($\chi^2 = 0.17$, $p = .68$), but a difference emerged for the recall, which was significant in the conventional analysis, but not once by-stimulus variability was accounted for in the mixed model ($\chi^2 = 1.2$, $p = .273$). To further examine this issue, we examined the effects of matching on individual messages. To this end, we averaged the relevant metrics on a by-billboard basis and computed correlations between measures, as well as individual examinations of attention and retention effects. As can be seen in Figure 4, there were fairly evident differences across message conditions. In particular, among the 20 experimental messages, some were recalled (and also recognized) very well, whereas others achieved low recall and substantially lower recognition. When computing various vector-correlations between recall and recognition metrics, we find all those correlations to be positive (although not all were statistically significant

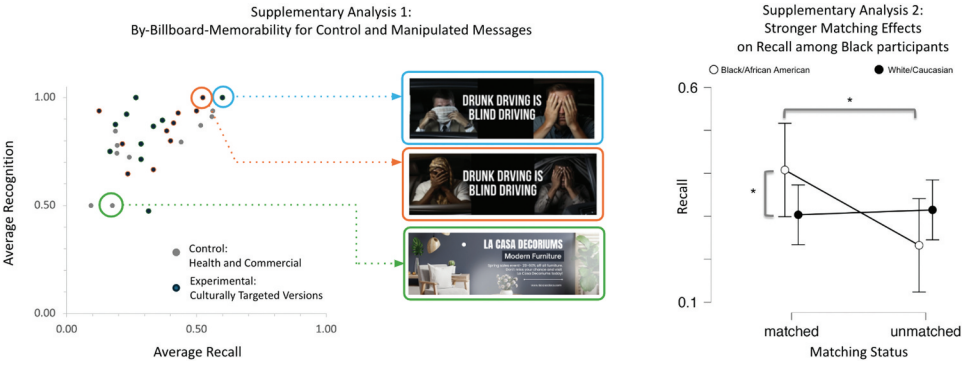



**Figure 4.** Illustration of selected results from exploratory analyses. Left panel: examining by-message (by-item/by-billboard) effects reveals robust correlations between individual billboards' memorability, i.e., their average recall and recognition rates. Moreover, it is clearly evident that some billboards were rather memorable – and others not. Right panel: In an exploratory analysis, we find an interaction between participants' race and the cultural targeting effect in terms of message recall. Post-hoc analyses (t-test) revealed that African American/black participants exhibit higher recall for culturally targeted (matched).

because of the low number of items per cell, i.e., 10 targeted, of which 5 were matched/unmatched, 5 neutral/untargeted health messages, 5 untargeted commercial messages). Similar results were obtained for the attention metrics (fixation count and total gaze duration). These results suggest that there are inherent differences between the billboards, which affect interest and memorability. Given that we invested significant effort in controlling the billboards' appearance and placement, we believe that these outcomes are due to topical factors (e.g., the alcohol messages being more relevant than, say furniture). Also, these results align with prior findings that presented by-message differences (Schmälzle et al., 2023). For instance, a billboard for a burger was among the most highly recalled messages in this study. We speculate that the advertisement for a burger restaurant was too distinct from other sets of health message billboards.

### *Analyses for participant subgroups (culture/race)*

Examining the subgroups of participants revealed an interesting pattern of effects. First, we found significant main effects for self-identified race on the metrics of visual attention, which were driven by White/Caucasian participants generally looking longer at billboards and revisiting more billboards (For Gaze Duration: $F(1,36) = 13.53$, $p < .001$; For Fixation Count: $F(1,36) = 6.5$, $p = .015$). A parallel pattern was observed for recognition, with White/Caucasian participants having higher recognition rates compared to African American/Blacks (*Recognition:* $F(1,36) = 5.7$, $p = .02$). For recall, the results revealed an interesting interaction between matching status and participant race (*Recall:* $F(1,36) = 4.6$, $p = .038$). In particular, as illustrated in Figure 4 (right panel) and confirmed by post-hoc t-tests (with Holm correction applied), the African American/Blacks group showed a stronger matching effect, recalling more matched than unmatched messages (Recall matched > unmatched: $t = 2.853$, $p_{holm} = .043$).

## Discussion

In this study, we examined the effects of culturally targeted messages on people's attention and memory via eye-tracking metrics within a virtual billboard environment. We found that billboards with images aligned with people's self-reported racial identity were slightly more likely to draw overt visual attention (measured via fixation count and gaze duration), though this was not statistically significant across all messages. Furthermore, we found that people recalled culturally matched messages more than unmatched messages.

First, we discuss the effects of using a design element of cultural targeting messages (i.e., surface-level matching with self-reported racial identity) on

visual attention. Our statistical analyses failed to identify strong effects of matching on visual attention. Specifically, we find that all of our participants fixated on each billboard at least once, suggesting a high base level of attention. Next, we zoomed in on the more nuanced attention metrics of gaze duration (how long they looked at the billboards in total) and fixation count (how often they revisited a given billboard). We found that both metrics do, in fact, show some evidence of more attention toward matched messages – gaze duration and fixation count were both nominally higher for the matched billboards compared to unmatched ones. However, this difference was not statistically significant. It remains speculative whether all of our message manipulations were explicit enough for participants to recognize the similarity of racial ingroup members portrayed on the billboards. In sum, these results do not support H1, although the by-message analyses (see Figure 4) suggested that the matching manipulation worked well for some messages.

Next, our second hypothesis stated that matched messages would lead to better retention. Our results provided support for this hypothesis, albeit with some qualifications. We did find that matched messages were recalled significantly more often, which supports H2. Put differently, this finding indicates that messages that matched a participant's racial identity were more frequently recalled than those that were not. This is consistent with other literature in message congruency showing that it can improve recall and result in continued recall over time (Lochbuehler et al., 2019). This is a promising finding for the design of culturally targeted or tailored messages as message recall is, and important part of the message reception process that can ultimately shape behavioral decision.

However, for the recognition measure of memory, we did not find corresponding retention benefits for message matching. It should be noted that recognition memory in general, and our specific version of the recognition test in particular, may not have been the best measure of retention in this context. While our recognition test displayed only the versions of the messages to which participants were exposed and additional distractors to account for general guessing tendencies, confusion or conflation might have happened between related topics (e.g., between drugged and drunk driving billboards). Also, the high level of successful recognition suggests that participants operated near the performance ceiling, which would also reduce the chance of finding differences between matched and unmatched billboards. Future studies might, for instance, increase the difficulty of recognition – be it through a longer retention interval, interfering memory tasks, or simply additional memoranda. With this in mind, we are inclined not to place too much weight on the recognition measure and instead focus on message recall. In memory research, it is well-known that recall is a more difficult task because it requires the active retrieval of a memory trace, and we do find that matched messages were recalled more often. This is in line with our predictions and thus provides support for H2.

The observed advantage of matched messages in memory (free recall), yet the lack of prioritized attention (gaze behavior) toward them during encoding, leads to the question as to when/where in the processing stream this effect arises. If all messages were looked at (attended to overtly) and there were no big differences in either the duration or the frequency of looking at matched vs. unmatched messages (i.e., no support for H1), this suggests that the improved recall for matched messages must originate from a post-perceptual processing boost. Moreover, this effect is slightly at odds with previous work that found prioritized overt attention (in terms of eye gaze) was directly related to improved retention (Hannula et al., 2007). It is plausible that differences in the experimental setup explain these seemingly incompatible effects; specifically, participants were instructed to perform a parallel, attention-consuming task and found that this task had a strong effect on overt attention -gaze behavior (e.g., Schmälzle et al., 2023). However, in the current study, only a single "free viewing" task in which participants were free to explore the highway and its surroundings (including the billboards) existed. As demonstrated by the fact that all participants looked at all billboards, this has led to a ceiling effect in terms of attention deployment toward the visual environment. Thus, given that we can empirically verify that all participants had viewed all billboards, we concluded that the recall-advantage for matched billboards must come into play through post-perceptual processes that take place after overt attention. For example, research on memory encoding and depth-of – processing has shown that memory – while obviously contingent on exposure and encoding times – is not just a function of exposure or encoding time (Craik & Lockhart, 1972; Craik & Tulving, 1975); rather, if participants are instructed to carry out, for example, semantic operations, even relatively short encoding times can procedure lasting memories – often more so than for items that were encoded a long time, but only with shallow operations. With this in mind, while participants looked at/attended to all items, it may be that they did implicitly categorize matching targets and that these sparked more post-perceptual engagement or elaboration processes.

The observed effect of culturally matched messages on participants' memory may be attributed to enhanced self-relevance, as suggested by Schmitz and Johnson (2007). When these culturally targeted messages are presented alongside culturally non-targeted messages, they might be more effective in promoting such risk perceptions as they are recalled better. This is also related to the own-race effect (ORE) in memory for faces, which demonstrates that people are inclined to remember faces of people who are perceived as their own race, compared to less familiar, or of other race faces (Brigham & Malpass, 1985; Meissner & Brigham, 2001; Zhou et al., 2021). However, this memory bias is contingent on an initial perceptual categorization; that is, the

effect is observed when the viewer first detects the stimuli as "matched" versus "unmatched" in their visual field (Prunty et al., 2023). Previous research on own-race face detection relies largely on reaction-time measures of accuracy and speed recognition (e.g., Barkowitz & Brigham, 1982; Wiese et al., 2014), although recent work using eye-tracking also exists (e.g., Burgund et al., 2024). Moreover, when it comes to understanding the mechanisms of own-race effect, other factors like attitudes and interracial contact might come into play (Meissner & Brigham, 2001) that go beyond the current data. Thus, while work on the own-race effect is relevant to the current study and research context, we believe more investigation is needed to integrate prior research with the current paradigm and cultural targeting framework.

Our supplementary analyses revealed additional insights worth mentioning. For instance, we found that different billboards varied in terms of the degree to which they were looked at and how often they were recalled, raising questions about the cause of these effects. Given that these were exploratory analyses, we can only speculate, but several potential answers come to mind: For instance, it could be that some topics are highly relevant – either in general or to many people (e.g., food > furniture); second, there could be visual features in the messages that contribute to saliency and memorability, two well-researched areas in visual cognition (Bainbridge, 2022; Itti, 2005) and research from the gestalt tradition would also suggest that the context can also influence memorability (e.g., one food commercial among 19 health commercials would likely stand out; Von Restorff, 1933). Although the current study randomized the presentation order, thereby ruling out order effects, the study relied on a fixed set of messages. Methodologically, this underscores the importance of modeling message-level variability in statistical analyses (e.g., Jackson & Jacobs, 1983) and raises several questions, such as whether our experimental matching manipulation was equally effective for all billboard exemplars as well as whether different health topics are more easily recalled. Generally, the between-message differences also align with prior work in which we found similar differences between more easy-to-recall topics (like buckling up) and more obscure topics.

### *Broader implications*

This research constitutes a new effort to investigate the mechanisms of cultural targeting, gauging message recipients' overt visual attention and retention in a virtual environment setting with high ecological validity compared to traditional laboratory experiments. Our findings corroborate the notion that targeting one's cultural identity with specific message content impacts message recognition and recall (Hale, 2022; Nelson & Garst, 2005). Moreover, we were able to observe that these effects may be intensified among specific groups, which aligns with the previous empirical findings on cultural targeting/tailoring messages that presented similar results (e.g., Campbell & Quintiliani, 2006).

Theoretically, this paper advances the newly described CTC framework (Lapinski & Oetzel, 2021; Lapinski et al., 2025) by providing empirical data regarding the effects of culturally targeted messages compared to non-targeted messages. The CTC framework distinguishes targeting from tailoring, separates cultural targeting/tailoring processes from content, identifies the elements along which tailoring can occur, and begins to theorize explanatory mechanisms for the effects of culturally congruent messages. In particular, we show that message retention, in particular, might mediate culturally targeted message effects.

In this paper, we further advance the understanding of Resnicow et al. (1999)'s distinction of surface- vs. deep-level of cultural targeting, and focus on a particular design element of cultural targeting by appealing to one's self-identified racial identity through messages; this can be classified as a widely used surface-level element (Lapinski et al., 2025). Prior studies which have examined responses to messages which varied the race of the people portrayed in messages but they often intentionally confound this design element with other message elements (Lucas et al., 2021; Manning et al., 2023); as such we know little about the unique effects of design elements. Social identity theory and self-categorization theory (Tajfel & Turner, 1986; Turner et al., 1987) deserve additional attention as the theoretical grounding to explain such effects. By demonstrating that culturally targeted messages resonating with one's valued social identity can have desired message outcomes, our results extend the applications of these foundational theories.

Furthermore, this study responds to calls in the cultural targeting literature by elucidating mediators that can explain the effects of these messages (Griffith et al., 2024). For instance, a recent review of culturally targeted/tailored messages notes a lack of theories that account for how and when culturally targeted/tailored message interventions should be most persuasive (Lapinski et al., 2025). According to Lapinski et al. (2025)'s review, although many studies target using design elements, few studies are able to extract the unique effects of these message elements because they vary multiple targeting elements. In our study, the messages were certainly targeted based on self-reported racial identity, but questions remain whether this manipulation led to increased identification as opposed to merely familiarity-based effects for the targeted messages. In any case, this all raises the point that studies often use different definitions and operationalizations of cultural targeting/tailoring, and suboptimal measurement methods to demonstrate that these manipulations are effective, i.e., lead to more attention, stronger identification, or whatever the theorized mechanism may be. Although our study cannot resolve these issues completely, the use of objective measures of eye-tracking to investigate attention as a first and necessary precondition for stronger targeting/tailoring effects points the way to improved ways of studying cultural targeting's mechanisms.

Moreover, this study adds an innovative methodological effort to the current cultural targeting/tailoring literature by operationalizing attention as part of the underlying process to explain message effects and gauging eye-tracking measures to empirically test such claims. While several existing research targets one's cultural identity to test the message effects of culturally targeted messages, most rely on self-report survey methods, which might be vulnerable to response bias or memory distortions. By contrast, the present study uses eye-tracking, which allows to objectively and unobtrusively track overt attention as a key element of the message reception process, which in turn helps explain message effects (King et al., 2019). As our results indicate, participants are more likely to be immersed in the research environment that is created by the researchers, which enhances perceived realism compared to traditional laboratory studies (Fox et al., 2009) yet maintains the strengths of lab-based research such as high controllability and effective message manipulations.

Regarding the practical implications of our findings, this research offers a bright future for message designers and public health communicators who are interested in testing the effectiveness of culturally targeted/tailored messages by applying emerging digital platforms and immersive technologies in diverse contexts. For instance, message communicators could collect more evidence to create engaging and memorable virtual billboards that resonate with specific cultural groups. Public health practitioners could utilize VR environments at relatively low cost to test and refine culturally targeted/tailored messages before implementing large-scale interventions, and potentially advance message modalities by applying them in different digital platforms such as Metaverse. Also, communicators could further benefit by subtly embedding cultural symbols and integrating billboard advertising into cohesive, culturally relevant campaigns across media platforms. Strategically placing culturally congruent billboards in relevant locations, dynamically adapting content on digital billboards to reflect real-time cultural contexts and targeting localized messages can amplify effectiveness. As communication interventions utilizing novel technologies and algorithms are becoming increasingly prevalent and advanced, even to the level of intricate personalization and tailoring, it is our hope that more communication scholars can attend to generating culturally resonant messages that can promote risk reduction behaviors in different contexts.

### *Strengths and limitations*

Although the current study demonstrates the effects of cultural targeting and uses a promising VR-based paradigm, some limitations should be acknowledged. First, while VR generally offers the possibility to create infinitely flexible but realistic and well-controlled environments, the current choice of

a highway driving context is only one singular instance of possible environments and may not maximize the potential of cultural targeting. Rather, it may be that targeting effects are more likely to be found in visually overloaded environments such as cities or other environments where the space is filled with "outgroup" information, which could make culturally matched billboards stand out more conspicuously (e.g., Wolfe & Horowitz, 2017). There could have been individual variability in terms of whether races is a metric for ingroup and outgroup perceptions, as the impact of ingroup exemplars on social identification and categorization can be weighted differently among individuals (Fazio & Dunton, 1997).

Another limitation of this study relates to the strength of the experimental manipulation. In particular, Figure 4 shows that while the billboards are very clearly matched to specific target groups, different versions of each feature a significant degree of overlap. Thus, the targeting manipulation is a fairly small subset of the entire information contained in the billboard. For example, billboard features we could also have manipulated include the text or other characteristics (like color scheme or cultural symbols). However, manipulating these characteristics via targeting would have added many visual confounds. Therefore, we opted for a tighter experimental control. Despite this limitation, the fact that we were still able to see targeting effects (nominally on the gaze metrics and significantly for the recall) speaks to the potential persuasive power of targeting.

Specifically, we argue that in the real world, where targeting manipulations are likely more subtle, the effects might accumulate to have a substantial impact. For example, in the current study, the recall benefit of matched messages was about 35% recall for matched messages compared to 30% for unmatched ones. On the face of it, this may seem like a relatively small advantage – and perhaps not all-too consequential with a laboratory sample of 40 participants. However, if we consider that an average highway billboard might be passed by about 10,000 cars per day, this will translate into 500 additional people who would actively recall the message. Although an exact number of daily billboard encounters is hard to obtain, reasonable estimates (e.g., based on daily number of drivers, length of drives, and numbers of billboard along roads) would suggest that every day, there occur about 5–10 billion instances of drivers (or vehicles) pass a visible billboard somewhere in the U.S. - and multiples of that number for the whole world. Given that active recall is a reasonably high standard for the impact of messages, it becomes clear how even apparently small effects may matter greatly in real-world impact (e.g., Percy & Rossiter, 1992), especially as effects accumulate over extended periods of time, like many memory-based media effects do (e.g., Shehata et al., 2021).

Another issue worth discussing is that of sample size and generalizability. We note that the sample size of 40 (20 per group) was technically large enough

to detect large- or medium-sized effects, but empirically the effects of the targeting manipulation were weaker than expected. This discrepancy between our expectation and observed effect size for the attention variable may have reduced statistical power and limited our ability to detect small effects. Therefore, the absence of a significant difference in attention (e.g., in H1) should be interpreted with caution. Future studies should consider studying larger samples as well as longer driving tasks to maximize the chance that even smaller effects could be more expressed (due to accumulation over time and multiple target objects (billboards) as well as over multiple drivers).

We also pose a connection between the effect size issue and the strength of our manipulation – as argued above with respect to how salient the represented models were and how the experimental context itself allowed such effects to become expressed. The study isolates one design feature of a message, racial model matching, from the broader set of cultural targeting strategies described in prior work (Lapinski et al., 2025), and thus the observed mechanisms (attention and recall) may not apply to other forms of cultural targeting or tailoring. Therefore, future research should expand our research by investigating whether similar or distinct mediating pathways emerge when other cultural targeting strategies (e.g., language, values, contextual visual cues) are manipulated.

Another piece of the puzzle is also the participants themselves, for instance, how salient cultural and racial issues were to them. In other words, such effects likely depend on the centrality of racial identity to a person's self-concept, which we did not measure, and for which our convenience sample was not optimized. Future research using larger and more generalizable samples should thus assess identity centrality as a potential moderator to clarify for whom cultural targeting is most effective, providing more nuance than broad demographic categories. Although our current lab-based VR sample has limited generalizability, we believe that as VR and metaverse-style communication systems continue to expand, it will become increasingly feasible to study topics like cultural targeting with large or very large samples – and potentially even including biometric measures like eye-tracking.

### *Avenues for future research*

This study provides several important directions for additional research. Especially as the metaverse develops, we might expect to encounter environments that can be preconfigured to maximize the overlap of environmental features with participants' cultural identities, preferences, and other proclivities. In fact, it is common to show participants specific feeds and other "matching" messages on social media platforms such as Facebook or YouTube. It is easy to see the parallels and opportunities (and threats) of

using this methodology for future communications campaigns. Going forward, it might be possible to match the environment (e.g., the room, type of highway) and the message content (i.e., the models shown) to participants' cultural and interest-based characteristics.

Theory-building regarding the ways in which cultural targeting impacts message response can only proceed with additional research on ways in which the various message elements impact message response through McGuire (2001)'s hierarchy of effects. Indeed, crossing the table of message elements described recently in Lapinski et al. (2025) with the hierarchy of effects could result in a large number of testable hypotheses about the many possible message modifications and their impact. Message elements could be targeted (as we have done in this study) or tailored to individual cultural characteristics. Because the message elements range from surface to deep-level cultural targeting, examining the effects of the range of elements will be useful for identifying the mechanisms of message effects with a goal of prediction.

Methodologically, we have only scratched the surface of the information that eye-tracking (and behavior tracking more broadly) offers in these new communication environments. In the current study, we used two eye-tracking metrics – gaze duration and fixation count – that represent fairly straightforward global metrics. For future studies, we could add the possibility of having specific areas-of-interest (AOIs) on the billboards (e.g., separate text and image information) rather than our current AOI as an entire billboard, or use various data-driven metrics and predictive modeling approaches to isolate the effective informational ingredients (e.g., human faces, fear/threat imagery) that drive experimental effect. Also, as both an avenue for future research but also a potential limitation of the current study, we note that the measurement of recognition could be expanded and revised, for instance by using forced-choice assessments between matched vs. unmatched exemplars, which would be a very clean comparison; likewise, to avoid interference between perceptual (billboard style), topical (e.g., smoking vs. risky driving), and the actual model (e.g., Black American vs. Caucasian), similar forced-choice comparisons could be used to develop potentially more sensitive recognition measures.

## *Summary and conclusion*

To summarize, we examined the effects of cultural targeting on participants' visual attention and memory using the VR-based billboard paradigm. It was revealed that matched messages are preferentially recalled but that attention did not vary based on match. As we enter the era of metaverse-mediated communication, it will become easy and likely commonplace to tailor content to recipient characteristics. From a communication intervention perspective, this raises new opportunities and ethical decisions as we

examine the effects of messages that use cultural targeting in many applied areas. We can expect that targeting will continue to be used as a strategy to influence people within risk, environment, and health communication-related areas. [9590 words]

## Notes

1. An *a priori* power analysis (G*Power 3.1) for a paired-samples t-test using a within-subjects design, indicated that a sample of $N = 27$ is required to detect a medium effect ($d = 0.5$) with 80% power at an alpha level of .05 (two-tailed) and $N = 12$ for a large effect ($d = 0.8$).
2. The ratio of the self-identified racial identity of the participants was 50:50 (i.e., White/Caucasian ($n = 19$) and Black/African American ($n = 19$). Three participants from the Black/African American group identified themselves as mixed-race.

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## Disclosure statement

No potential conflict of interest was reported by the author(s).

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